Bias in Bios: Project 1 Quick Guide

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We guess a person's occupation (28 options) from their bio, compare 9 systems with cross-validation and a Bayesian test, and check fairness between women and men. Best: MLP + TF-IDF, macro-F1 0.754. Every system amplifies gender imbalance (r = 0.86). The full version is the Project 1 Study Guide.

Numbers to remember

What	Number
Bios (train, after removing duplicates)	257,081; 28 occupations
Imbalance	professor 29.8%, rapper 0.35% (84×)
"Always professor" baseline	accuracy 0.298, macro-F1 0.016
Women per occupation	9.7% (rapper) to 92.9% (dietitian); 46.1% overall
Pronouns reveal the gender label	95% of bios, 100% agreement
Tokenizer	own BPE, 10,000 tokens; 21/28 job names = 1 token
Cross-validation	3 folds × 2 seeds = 6 scores per system
Best system	MLP + TF-IDF: 0.754 ± 0.002 macro-F1 (accuracy 0.811)
Model averages	MLP 0.750 · deeper MLP 0.745 · log-linear 0.732
ROPE	±0.01 macro-F1 (≈ 3× fold noise of 0.0031)
Gender gap vs % women	r = 0.86 (all systems 0.83–0.90)
Biggest gaps	model +0.49, surgeon −0.21
Pronouns deleted	macro-F1 0.754 → 0.751; RMS gap 0.149 → 0.116 (−22%)

The six questions, one breath each

1. **Dataset.** 257k English web bios; the sentence naming the job is removed; 28 classes, very imbalanced (professor 30%). Gender is binary, inferred from pronouns, used only to check fairness. We removed 15 duplicated bios with conflicting labels.

2. **Tokenizer.** Our own BPE, trained on our bios (257k is plenty), 10,000 tokens, merges only for pairs seen ⩾ 5 times. Lowercased, punctuation split off. Not the tutorial's AG News tokenizer: job words would be fragmented.
3. **Features.** Indicator (is the token there?), counts (how often?) and our TF-IDF (damp repeats, down-weight common words, remove length; idf learned on the training fold only). 78% of counts are 1, so indicator ≈ counts.
4. **Architectures.** Log-linear (clues add up), MLP with 128 hidden units (combinations), and our deeper MLP: a second layer added through a shortcut, with LayerNorm. Its first version lost to the MLP; the redesign matches it. Early stopping on a 10% hold-out from the training part; one setting tuned per system.
5. **Evaluation.** Macro-F1, because accuracy rewards ignoring rare jobs. Stratified 3-fold cross-validation × 2 seeds, the same splits for all systems. Fairness: recall of women minus recall of men per job (TPR gap) and its RMS.
6. **Best system.** MLP + TF-IDF (0.754), but log-linear + TF-IDF (0.749) is practically equal by the Bayesian test (P = 0.998). One hidden layer helps unless TF-IDF is used; a second layer adds nothing; indicator = counts.

Likely follow-ups, one line each

Question	Answer
Why macro-F1?	Each of 28 jobs counts equally; "always professor" = 0.30 accuracy but 0.016 macro-F1.
Precision / recall / F1?	Right when we say *y* / found of all real *y* / 2PR/(P+R).
Why cross-validation?	One split gives one random score; 6 scores show the variation the Bayesian test needs.
Why stratify?	Every fold gets the same job mix and % women (rare jobs, 88 female rappers).
Why 3 × 2, not 5 × 3?	25 min instead of 1.5 h; cost: wider posteriors.
Is the tokenizer leakage?	No labels involved, same for all systems; strict option: train it on dev.
Bigger vocabulary better?	Not always: more meaningful units, but rarer features and more weights.
Man bites dog?	All three representations are bags: word order is lost.

Question	Answer
Why TF-IDF helps log-linear but not the MLP?	The MLP learns its own weight per word; log-linear's penalty shrinks all weights equally.
ReLU? Dropout?	ReLU = max(0, z), makes layers non-linear; dropout switches off random units in training.
Why early stopping on a training hold-out?	The tutorial picks the epoch on the validation fold and reports it: optimistic.
Why did the deeper MLP need a redesign?	TF-IDF's tiny numbers starved layer 2; LayerNorm and a shortcut fixed it.
Posterior / ROPE / HDI?	Belief about the true difference / "too small to matter" band / narrowest 95% range.
Why "correlated" t-test?	Folds share training data, so scores aren't independent; the variance is inflated.
Why a ROPE of 0.01?	A difference we'd act on, ≈ 3× fold noise, checked with 0.005 and 0.02.
Why not a p-value?	It only asks "exactly zero?" and hides size; estimation can say "practically equal".
TPR gap / RMS gap?	Recall(women) − recall(men) per job / √(mean gap²): one fairness number.
Where do missed bios go?	To a related, gender-typical job: male paralegal → attorney, female surgeon → physician.
Does deleting pronouns fix bias?	No: gaps −22%, but names and words like *husband* still give gender away.

Weak spots to admit

- Gender is binary and inferred from pronouns, a proxy.
- Only 6 scores per system, so posteriors are wide.
- One setting tuned per system, on one fold; the deeper MLP's tricks were checked on one split only.
- Log-linear on counts did not fully converge.
- Near-duplicate templates can sit in both training and test parts.
- Pronoun deletion leaves first names and other proxies.
- dev and test sets not used yet.